Mining Dairy Data with AI Agents

Pádraig Lucey

What is an agent?

[An automation] that runs tools in a loop to achieve a goal.

Simon Willison

'AI' is not a tool.

Rich Hickey

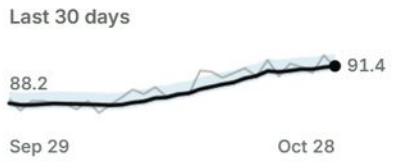
Goal

Average ECM per cow in pen 21?

- ✓ Reading the herd records `load_herd()` **6,980 cows**
- ✓ Finding pen 21's milking cows `filter(pen = 21)` **238 cows**
- ✓ Working out ECM `ecm(milk, fat, protein)` **91.4 lb**

91.4 lb ECM a cow

238 cows



Goal

Tell me about pen 21’s intake over the last month, what’s changed, and how the cows are responding.

Goal

Tell me about pen 21's intake over the last month, what's changed, and how the cows are responding.

- ✓ Reading dry matter intake `read_intake(pen = 21, days = 30)` **30 days**
- ✓ Pulling recipes `read_recipes(pen = 21)` **Sep 1 · today**
- ✓ Searching your email for forage tests `search_email("corn silage")` **2 lab reports**

Three changes on Sep 28

- ⊙ A new corn silage pile, with its lab report
- ＋ Half a pound more bypass fat
- － Half a pound less rolled corn

Goal

Tell me about pen 21's intake over the last month, what's changed, and how the cows are responding.

Worked for 4 s · 3 steps

- New corn silage pile
- + ½ lb more bypass fat
- − ½ lb less rolled corn

Recipe · High cows

Sep 1

Dry matter intake	60.0 lb
Corn silage old pile	16.0
Alfalfa hay	8.5
Wheat silage	2.5
Almond hulls	3.5
Rolled corn	12.5
Whole cottonseed	4.0
Canola meal	6.0
Distillers grains	2.0
Soybean meal	2.5
Molasses	0.5
Bypass fat	0.5
Minerals and buffer	1.5

Nutrients, % of DM

NEL	0.75 Mcal/lb
CP	16.5%
aNDF	29.9%
Forage NDF	18.5%
Starch	25.0%
Sugar	5.3%
Fatty acids	4.3%

Recipe · High cows

Today, Oct 28

Dry matter intake	63.0 lb +3.0
Corn silage new crop	~~16.0~~ 17.5
Alfalfa hay	9.0
Wheat silage	2.5
Almond hulls	3.5
Rolled corn	~~12.5~~ 12.0
Whole cottonseed	4.0
Canola meal	7.0
Distillers grains	2.0
Soybean meal	2.5
Molasses	0.5
Bypass fat	~~0.5~~ 1.0
Minerals and buffer	1.5

Nutrients, % of DM

NEL	0.75 Mcal/lb
CP	16.6%
aNDF	30.5%
Forage NDF	19.4%
Starch	23.7% −1.3
Sugar	5.3%
Fatty acids	4.9% +0.6

Goal

Tell me about pen 21's intake over the last month, what's changed, and how the cows are responding.

Worked for 4 s · 3 steps | New corn silage pile | + ½ lb more bypass fat | − ½ lb less rolled corn

✓ Opening pen 21 `open_pen(21)` **75 days**

Pen 21

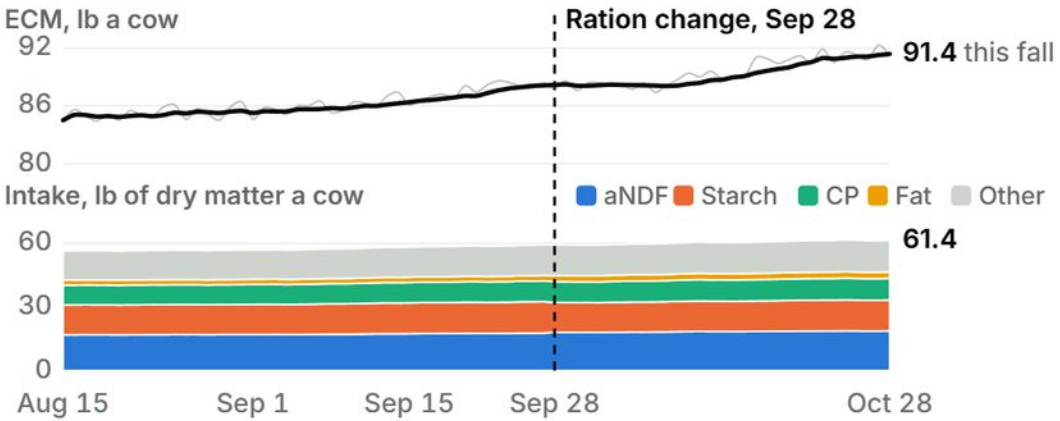
238 cows · High cows · Aug 15 to Oct 28

Milk a cow, last test
90.6 lb Highest of 12 pens
Middle 80%: 73–108 lb

Days in milk
52 median
Middle 80%: 31–74 days

Lactation
3rd median
2nd 84 · 3rd 51 · 4th+ 103

Intake, last 7 days
61.4 lb formulated 63.0
Daily, the last 30 days



Goal

Tell me about pen 21's intake over the last month, what's changed, and how the cows are responding.

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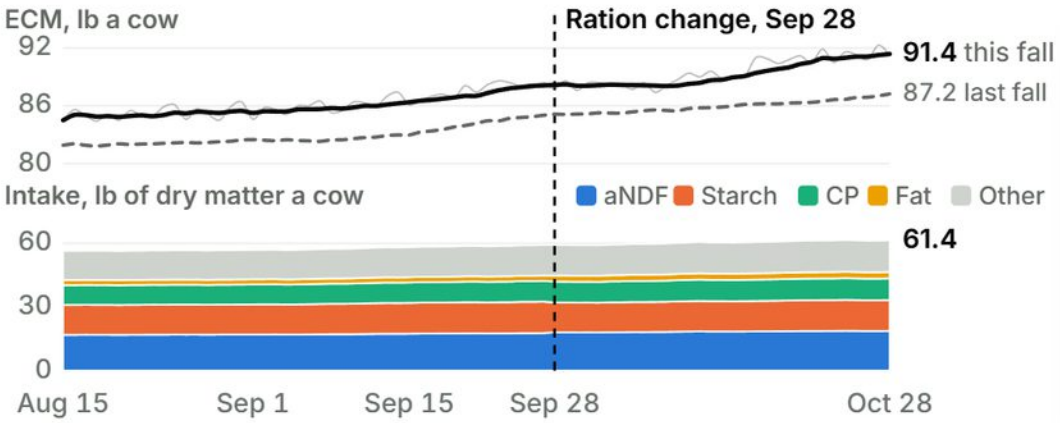
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61.4 lb formulated 63.0
Daily, the last 30 days



Maybe this change is just seasonal?

Checking the same weeks last fall.

ECM is up 3.6 lb since Sep 28. Last fall, same weeks, no ration change: up 1.9 lb.

Goal

Tell me about pen 21's intake over the last month, what's changed, and how the cows are responding.

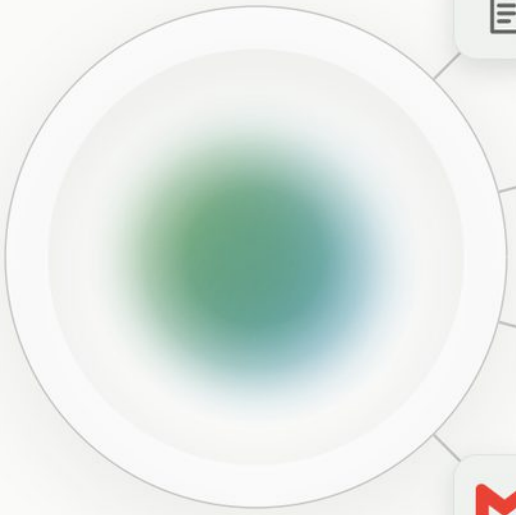
Model, harness, agent

The model

Reads your prompt and predicts the words that come next.

The harness

Gives the model its tools, and runs them.



Tools

Read a data file
↩ Cow file and feed report

Weather station API
↩ Temperatures, both weeks

Run statistics
↩ A difference from Sep 28

Email MCP
↩ 2 lab reports

After Harness Engineering 101, Isuru Wijesiri, 2026

AGENTS

AGENTS

Nutrition · Herd A · close-up pen

No Close Up Mineral in the 13 Oct premix

Close Up Mineral	**0** of 1,410 lb
Fed to the close-up pen	**14–16** Oct
The 12 batches before	1,450–2,110 lb



HOW WE KNOW

The wagon scale gained 0 lb during the Close Up Mineral step. EZfeed loading records.

Close-up minerals often carry the anionic salts that lower milk fever risk.
Santos et al., J Dairy Sci 2019

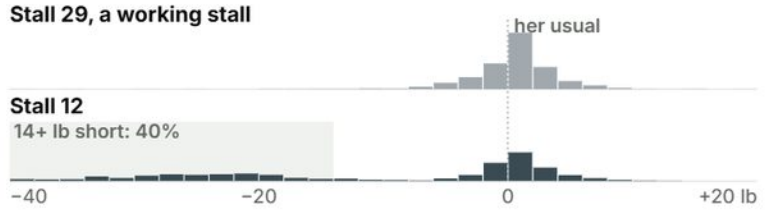
Did the 13 Oct batch get its Close Up Mineral?

Yes | No | Don't know

Milk · Herd B · rotary parlor

3 stalls milk low

Stall 12	**−10.2** lb a milking
Stall 57	**−8.9** lb
Stall 73	**−7.8** lb
The other 77 stalls	−2.9 to +2.4 lb



HOW WE KNOW

40% of stall 12's milkings came in 14 lb or more short of the cow's usual; 1% in a working stall.

Meter or milking unit? A meter reading short loses nothing; a unit not milking out leaves about 1,300 lb a day in the cows.
Andreen et al., J Dairy Sci 2020

Have stalls 12, 57 and 73 been checked?

Yes | Not yet | Don't know

AGENTS

Two different harnesses

🧩 **Mini Crossword**

Fill in the squares using the clues below. Each number marks the beginning of a word.

Plain text

1	2	3
4		
5		

➡️ **Across**

1. A small feline pet (3 letters)
4. To exist, as in "you ___ here" (3 letters)
5. The number after nine (3 letters)

⬇️ **Down**

1. A small feline pet (3 letters)
2. A word meaning "to exist", used with "you" (3 letters)
3. The number after nine (3 letters)

Here's the completed crossword with the answers filled in. 🧩

Plain text

C	A	T
E	S	T
T	E	N

Across

1. CAT 🐱
2. EST (short for "exists")
3. TEN

Down

1. CET
2. ASE
3. TTN

Note: The original puzzle's down clues don't produce valid English words, so this isn't a properly constructed crossword. I can make you a proper crossword with interlocking words and a complete, correct solution instead!

August 2026

Riemann's 160-year-old problem: two-thirds proven

Claude proved more than two-thirds of the zeta zeros lie on the line: 60 sub-agents, a day and a half

The first 15 zeros of the zeta function, on the critical line



Share of all its zeros proven to lie on the line



@alexanderrX_ on X · 19 Oct 2026

Claude, with L. Alpöge and R. Furman · arXiv 2608.13637 · Aug 2026

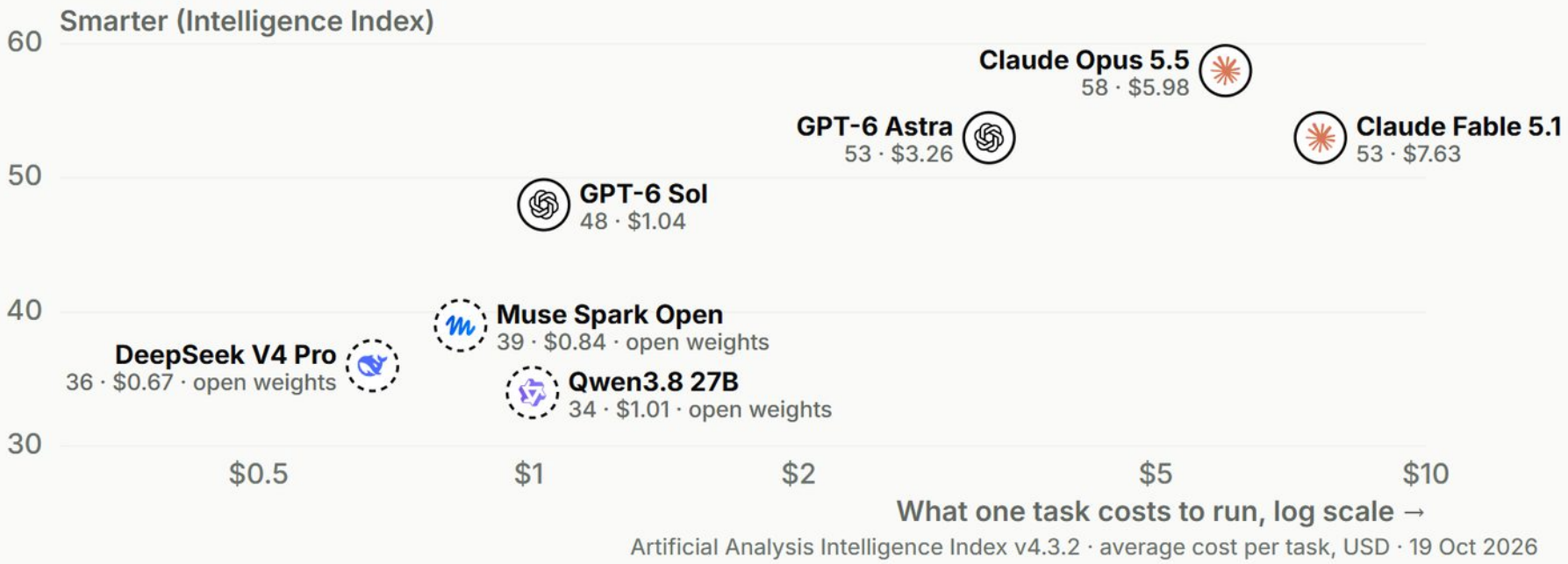
MODELS

MODELS

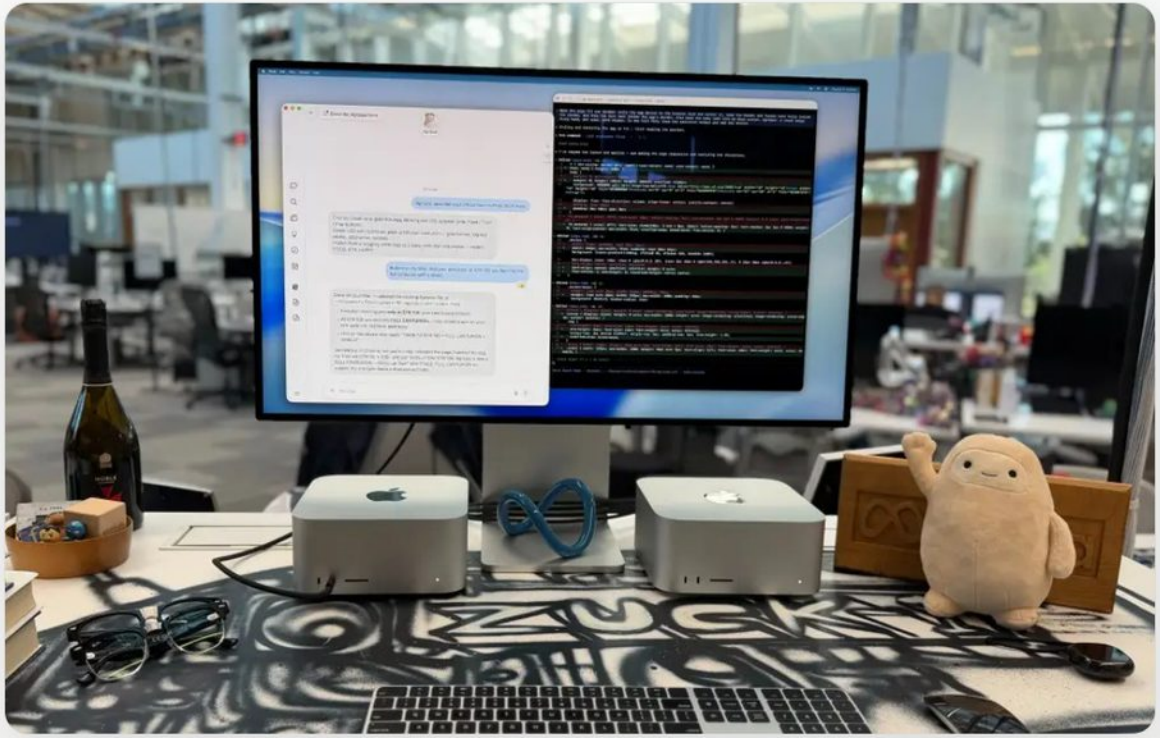
Cloud access or locally hosted

CLOUD ACCESS, ON THEIR SERVERS

LOCALLY HOSTED, ON YOUR MACHINE



MODELS

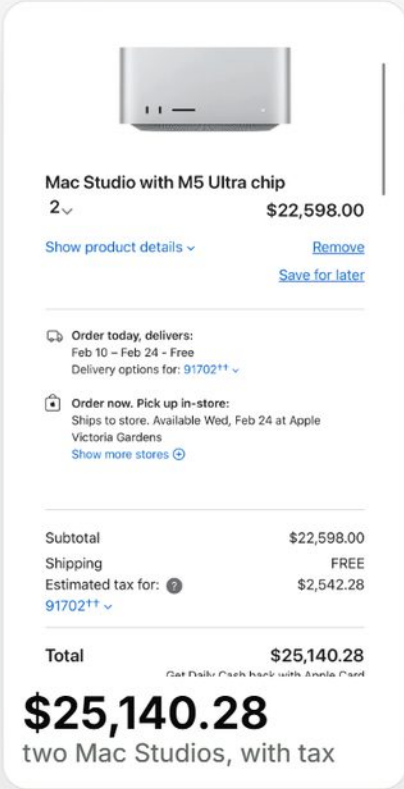


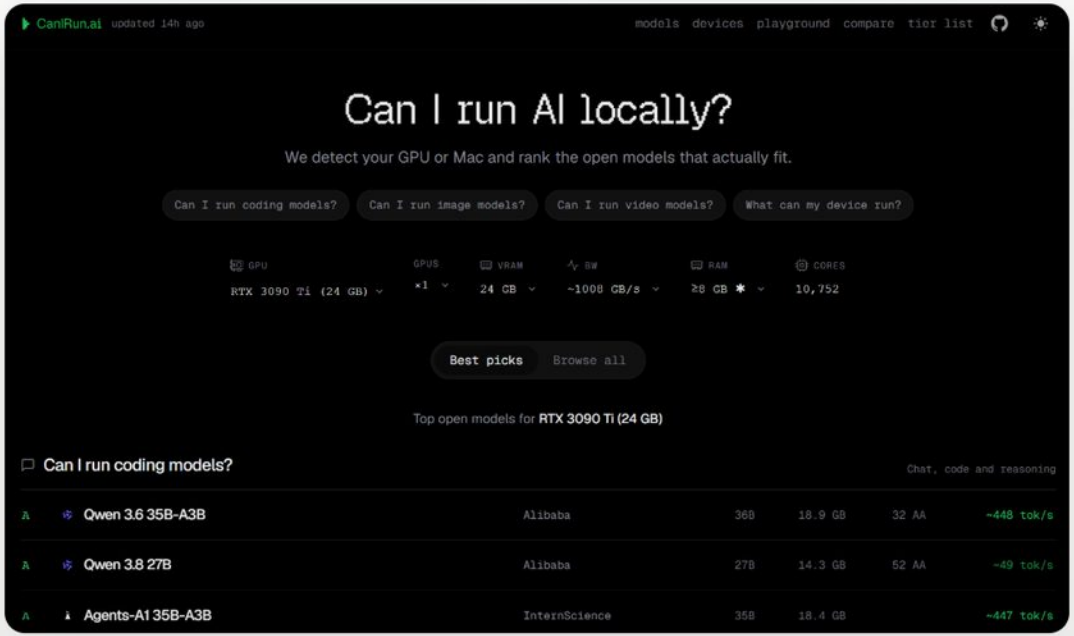
Muse Spark Open
Meta, 15 Oct 2026

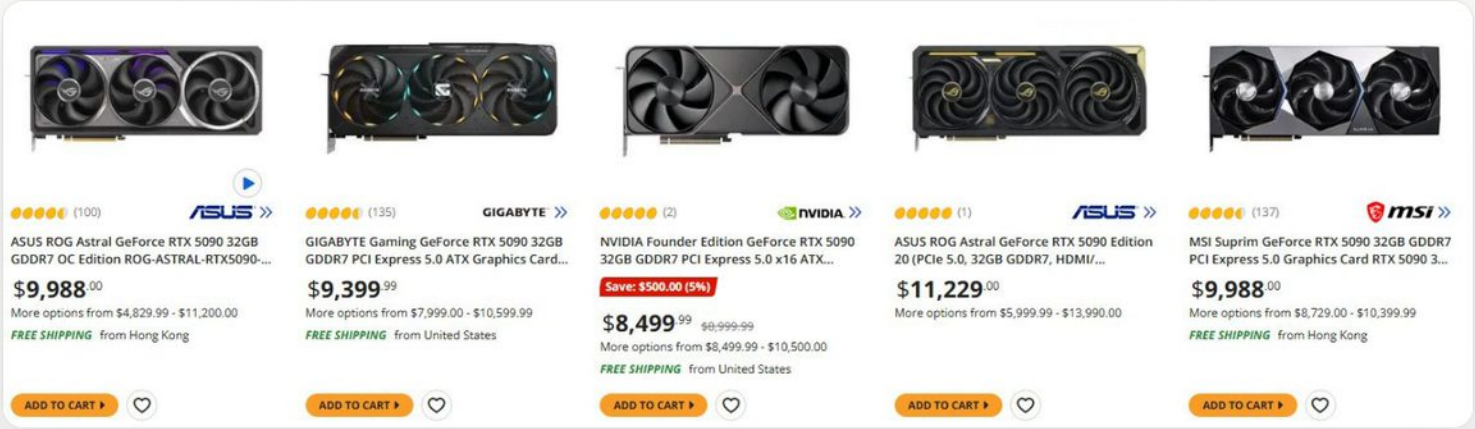
816B parameters

Two Mac Studios
512 GB of memory each

16–37
tokens a second







canirun.ai, for an RTX 3090 Ti (24 GB) · Newegg, RTX 5090, Oct 2026

RStudio · Herd project

fresh-cows.qmd

```
1. ---
2. title: "Herd project: fresh cows"
3. format: html
4. ---
5.
6. ## The fresh pen
7.
8. Which diseases, and when?
9.
10. ```{r}
11.
12. ```
```

Console

```
> library(vmtrc)
Loading the assistant's tools
  ✓ model       Qwen3.6-35B-A3B
  ✓ notes       epi, nutrition, rations, R help
  ✓ functions   tested course R
  ✓ benchmarks  37 herd benchmarks
  ✓ checks      every number
  ✓ library     CDC, USDA NAHMS
  ✓ web search  no farm details
>
```

Assistant Qwen3.6 35B-A3B

Ask about your herd, your code or the course

Message the assistant

Assistant

Qwen3.6 35B-A3B

Ask about your herd, your code or the course

How do I open my DairyComp files and look at fresh-cow diseases by days in milk?

Assistant

Qwen3.6 35B-A3B

```
fresh <- c("milk_fever", "retained_placenta", "metritis",
           "ketosis", "displaced_abomasum")
cases <- subset(herd$events, std_event %in% fresh & dim_at_event <= 60)
cases$dim <- cut(cases$dim_at_event, c(-1, 7, 14, 21, 30, 60),
                 labels = c("0-7", "8-14", "15-21", "22-30", "31-60"))
table(cases$std_event, cases$dim)
```

Output

	0-7	8-14	15-21	22-30	31-60
displaced_abomasum	0	6	8	7	0
ketosis	13	30	27	0	0
metritis	27	61	0	0	0
milk_fever	17	0	0	0	0
retained_placenta	55	0	0	0	0

Milk fever and retained placenta are all in the first week; metritis peaks in week two, ketosis in weeks two and three, and displaced abomasum runs from week two to four.

Insert into my file | Run it | Think harder

Message the assistant

HARNESS

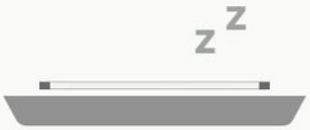
HARNESS

Where it runs

5:00 am

HARNESS Installed on the machine or In a sandbox

MODEL On the machine, like the Rossow lab or A frontier model, through its API



Your laptop

Only while it's open

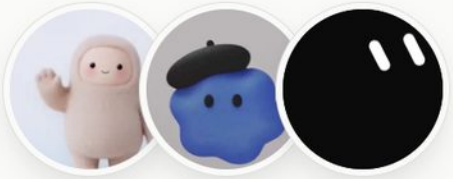
○ Lid shut, asleep



A box in the office

A Mac mini, or a $5 Ubuntu server

● Running



Their cloud

Muse, dots, Grok Bot

● Running

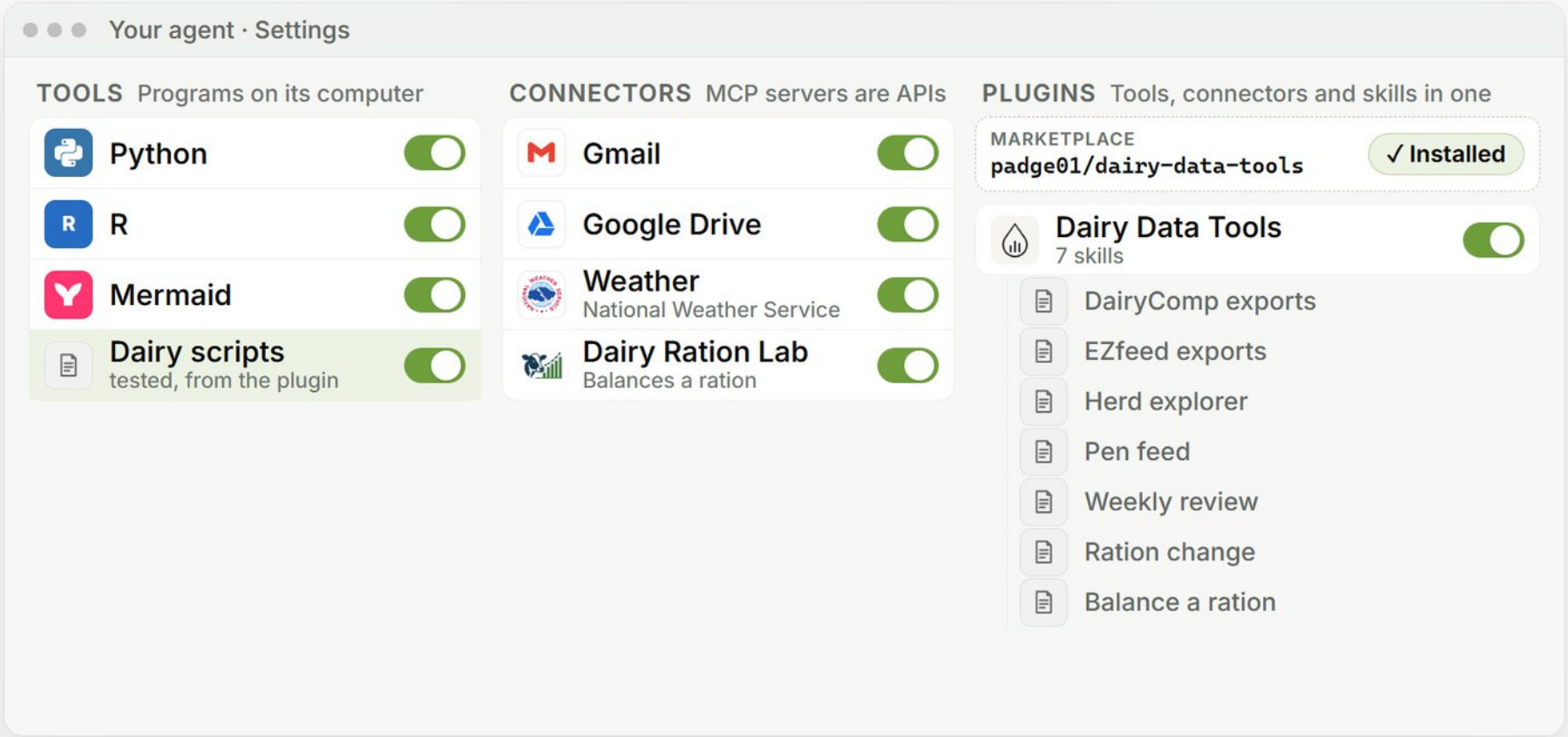
FOUND AT 5:00

NUTRITION New haylage sample added to your feed library

MILK Bulk tank SCC up from 180,000 to 260,000 this week

HARNESS

Tools, connectors and plugins



TOOLS

TOOLS

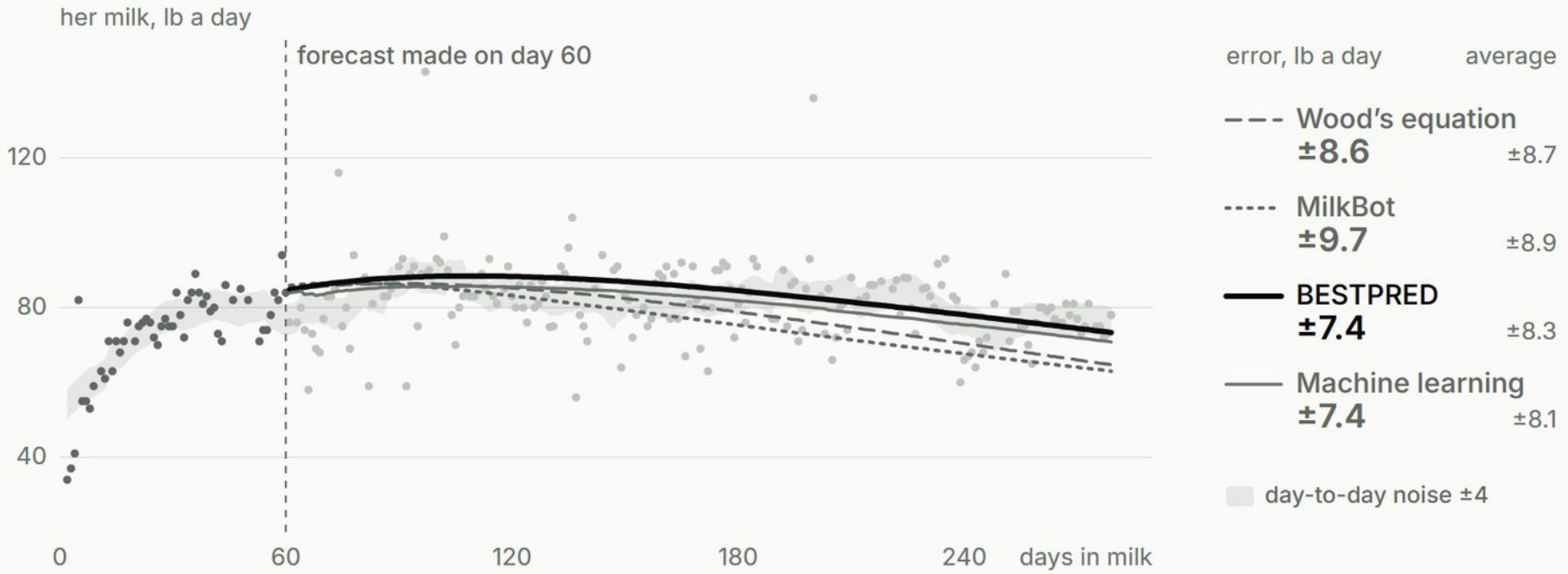
BESTPRED

```
$ claude plugin marketplace add padge01/bestpred-python
```

```
$ pip install bestpred
```

Python port of the USDA lactation curve model

Her test days (milk, fat, protein) · calving date · lactation · age · breed



BESTPRED: Cole and VanRaden, USDA-ARS Animal Genomics and Improvement Laboratory (2008) · Python port: doi.org/10.5281/zenodo.23425506

TOOLS

Dairy Data Tools

```
$ claude plugin marketplace add padge01/dairy-data-tools
```

```
$ pip install dairy-data-tools
```

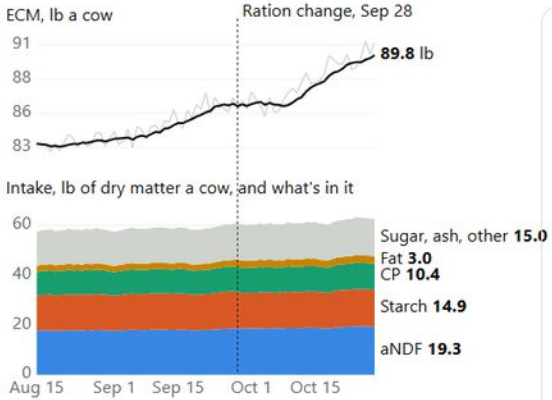
Pen 21 129 cows · Milking cows · Aug 15 to Oct 28

Milk per cow, last test
86.0 lb ECM 89.8 lb
Highest of 3 pens · middle 80%: 56–119 lb

Days in milk
173 median
Middle 80%: 62–320 days

Lactation
3 median
2nd 47 · 3rd+ 82

Intake, last 7 days
63.0 lb DM per cow
Called for 62.9 lb · daily, last 30 days



Ration, last 7 days

lb of dry matter per cow, called for

FORAGES
- Corn silage, 2026 pile — 17.0 lb
- Alfalfa hay — 8.7 lb
- Wheat silage — 2.4 lb

MINERALS AND ADDITIVES
- Lactation mineral — 1.5 lb

FATS
- Bypass fat — 1.0 lb

BY-PRODUCTS
- Whole cottonseed — 3.9 lb
- Almond hulls — 3.4 lb

PROTEINS
- Canola meal — 6.8 lb
- Soybean meal — 2.4 lb
- Distillers grains — 1.9 lb

GRAINS AND ENERGY
- Rolled corn — 11.7 lb
- Molasses — 0.5 lb

Cow 3256 Pregnant 147 d

Pen 21 · Holstein · 5th lact

77 lb, last test **245** DIM

Bottom 40% of pen 21, last test (129 cows)

Her milk against pen 21, last test (lb)



Lowest 16 · her 77 · highest 162

Her milk this lactation, lb by days in milk (each test)



Fat · protein	ECM, last test	SCC
3.9% · 3.6%	85 lb	138k

Jul 30	Footbath
Jul 29	Lameness
Jul 6	Checked pregnant
Jun 3	Bred

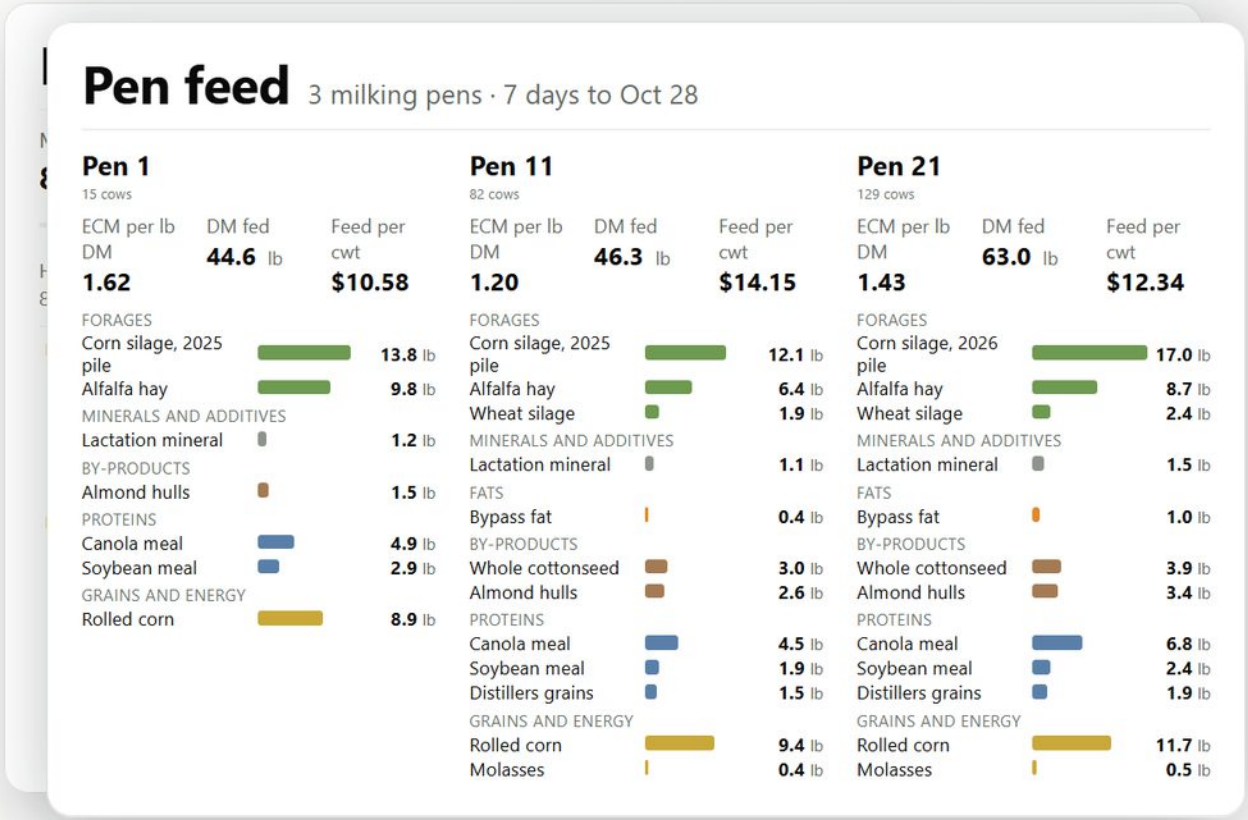
+4 earlier this lactation

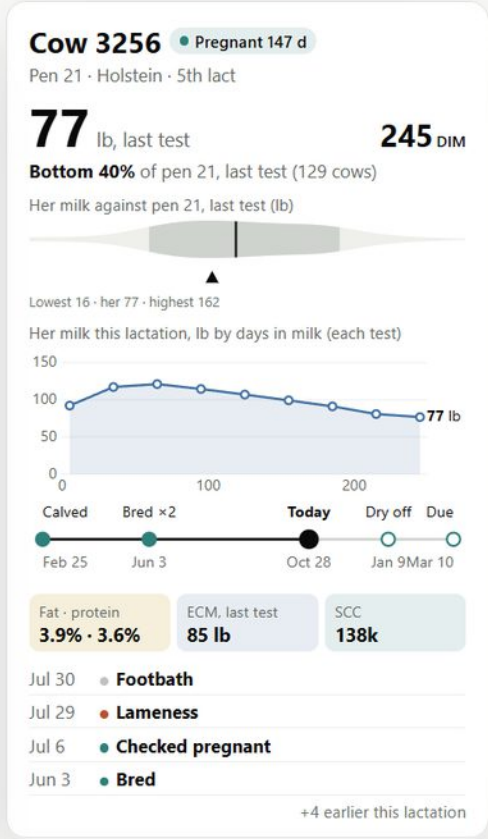
TOOLS

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```

```
$ pip install dairy-data-tools
```





MCP AND API

MCP AND API

Dairy Ration Lab

```
$ claude mcp add --transport http ration-lab \
  https://mcp.dairydatascience.com
```



DAIRY HERD/ NEWS/ EDUCATION

Should AI Be Formulating Dairy Rations? Experts Say Not Yet

Artificial intelligence is becoming a powerful tool for organizing and analyzing dairy data, but nutritionists say it still is not ready to take over one of the most important decisions on the farm.

By **Taylor Leach** · May 27, 2026 10:06 AM

AN MCP SERVER

The software's list of functions

Written as instructions for AI

Your AI runs them, signed in as you

`formulate` `search_feeds` `worth_buying` `read_ration` `set_price` `load_sheet`

Your AI agent — `MCP` — **Dairy Ration Lab**

Formulate a ration on its own? **No.**

Run the ration software for you? **Yes.**

MCP AND API

Dairy Ration Lab

```
$ claude mcp add --transport http ration-lab \
    https://mcp.dairydatascience.com
```

Balance a least-cost ration for my High pen, inside my constraints: 94 lb of milk, 89 days in milk, eating 60 lb.

RATION-LAB **open_example_farm**

RATION-LAB **set_price** 10 feeds, USDA California

RATION-LAB **formulate** High, least_cost

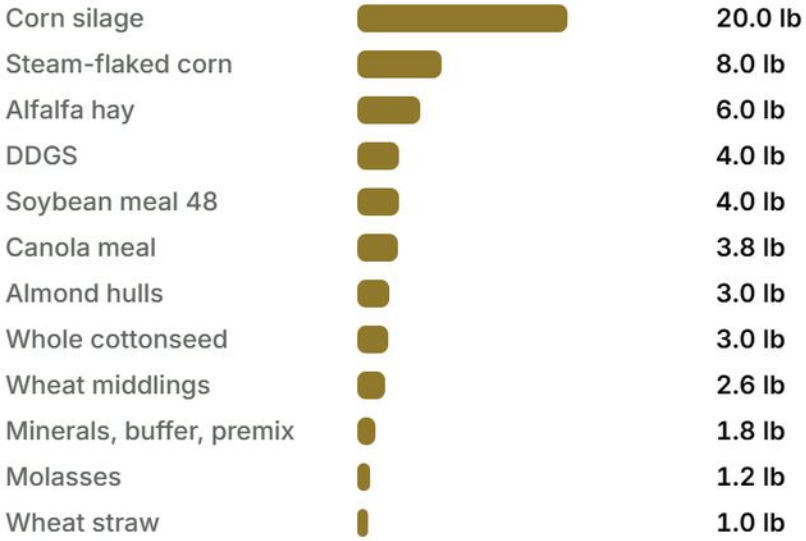
Least cost for High inside your constraints, at this week's California prices: $8.58 a cow a day on 58.5 lb of dry matter, IOFC $6.80. Energy and protein met.

Ration Lab High · least cost · your constraints — Balanced

58.5 lb dry matter

$8.58 a cow a day IOFC $6.80

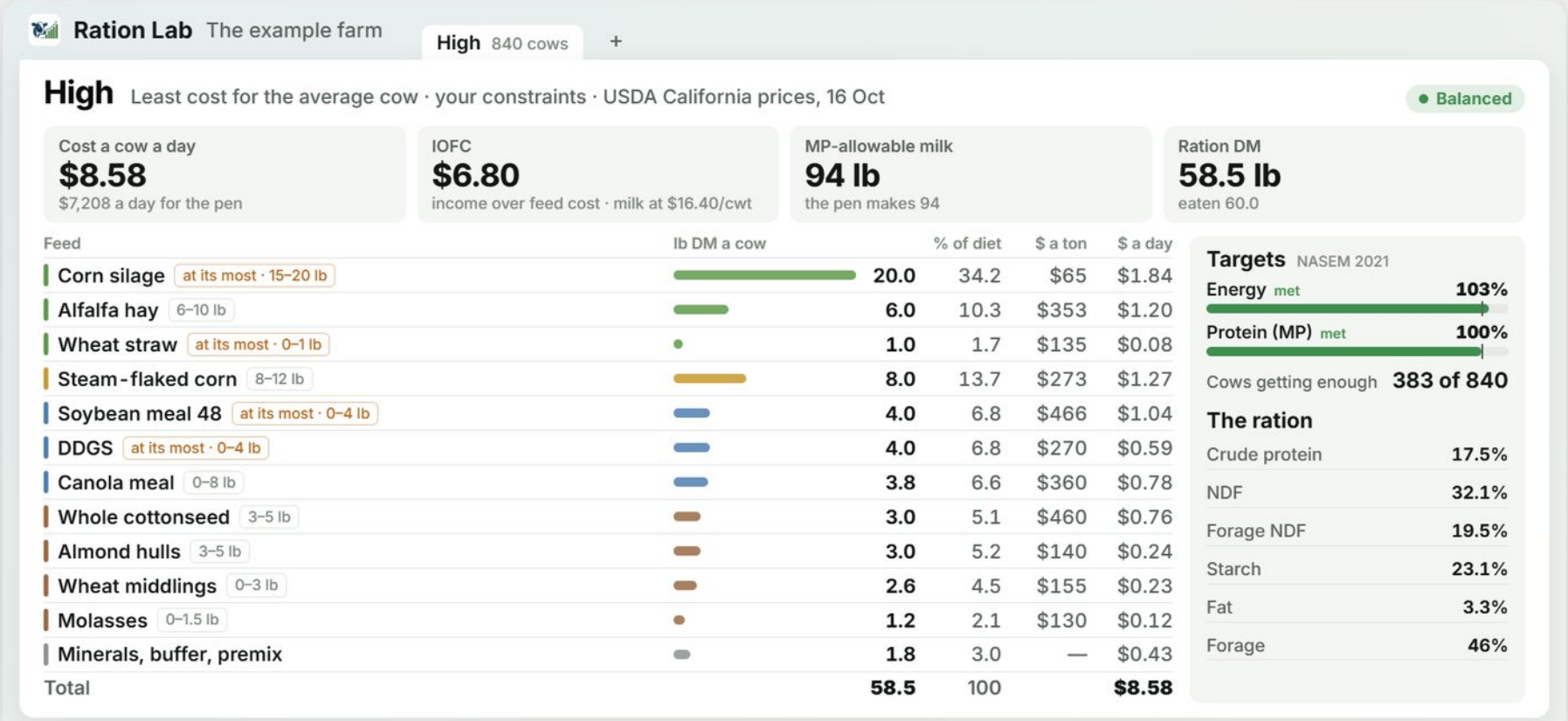
Your constraints · Feed: USDA California prices, 16 Oct 2026 · Milk: USDA September component prices, $16.40/cwt



MCP AND API

Dairy Ration Lab

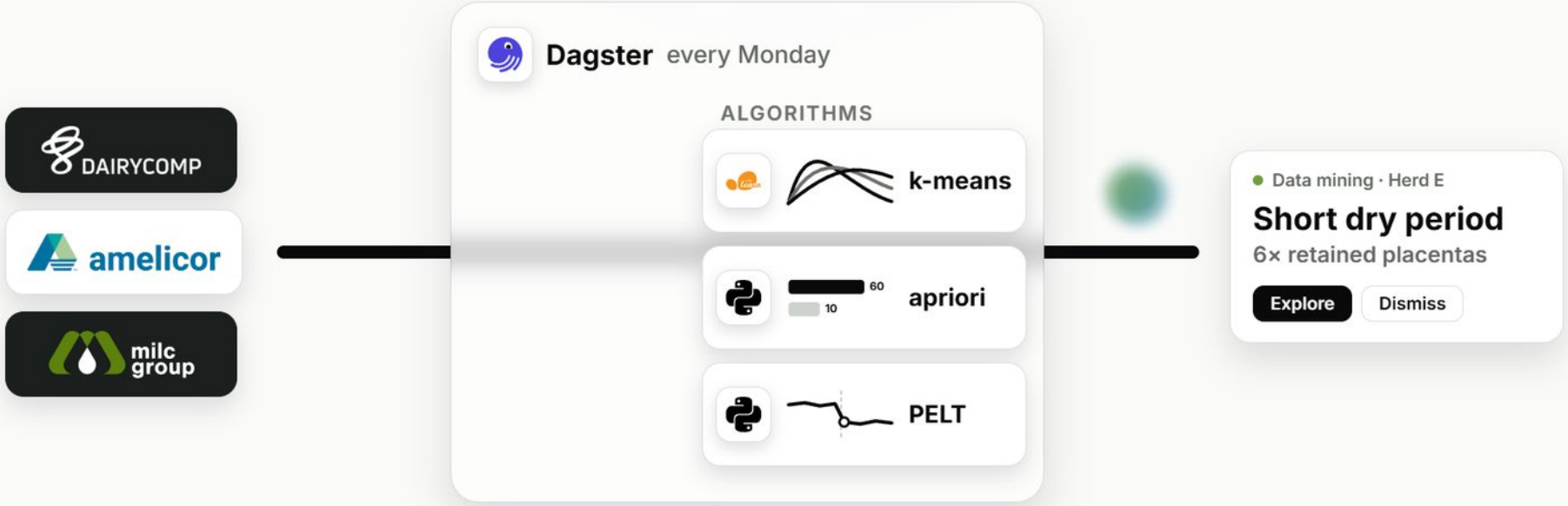
```
$ claude mcp add --transport http ration-lab \
  https://mcp.dairydatascience.com
```



DATA MINING

DATA MINING

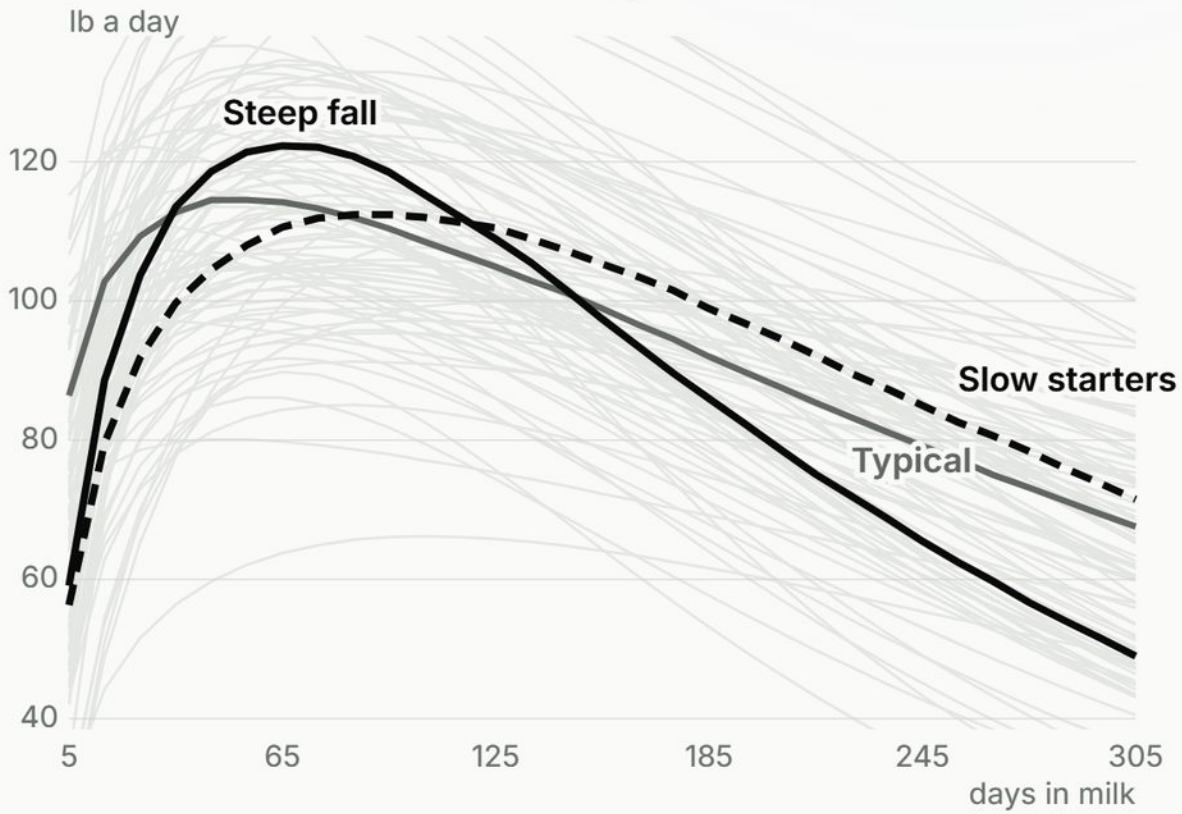
Agent-run ML pipelines



DATA MINING

Three shapes of lactation

The median curve of each shape



NO LABELS GIVEN

2,706 lactations, sorted only by the shape of the curve

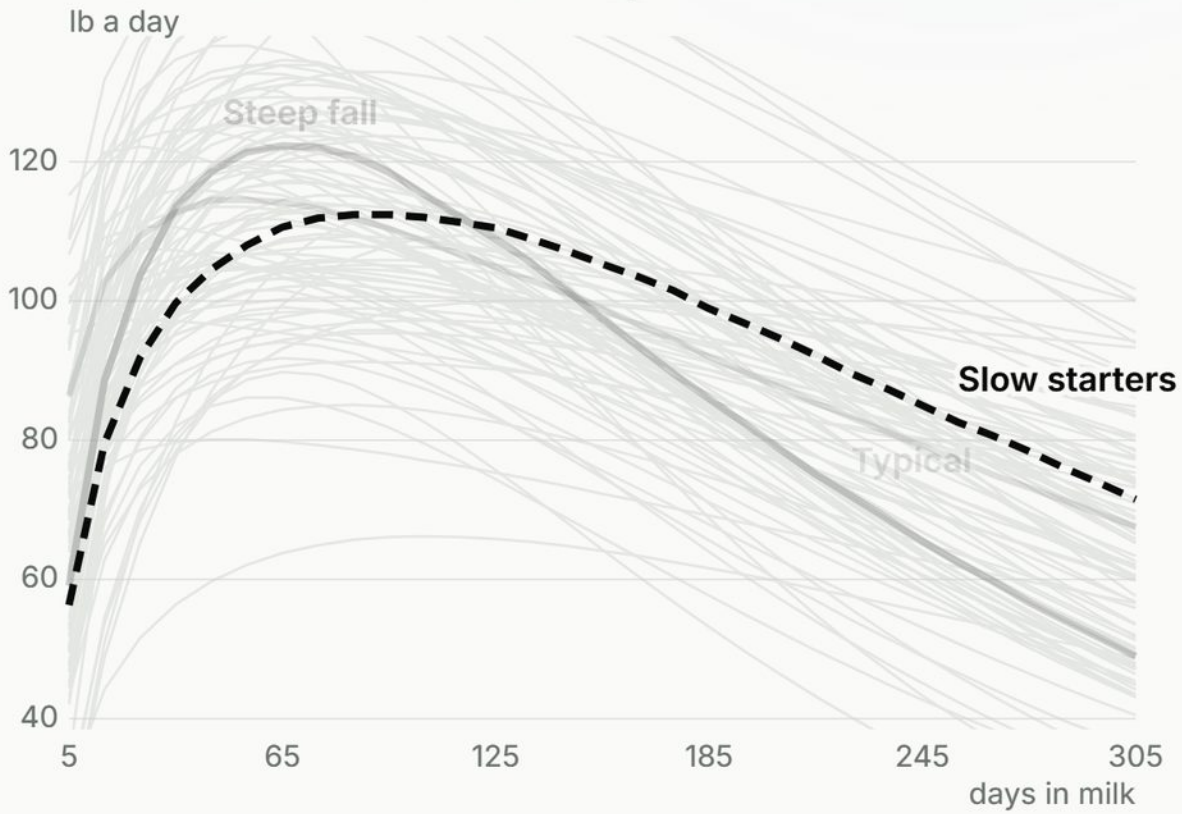
peak day · rise · persistency
90 of them drawn here

```
k-means, 3 groups
2,706 lactations · 6 dairies
```

DATA MINING

Three shapes of lactation

The median curve of each shape



SLOW STARTERS

Cows with a retained placenta
285 cows

35 in 100

Cows without a retained placenta
1,892 cows

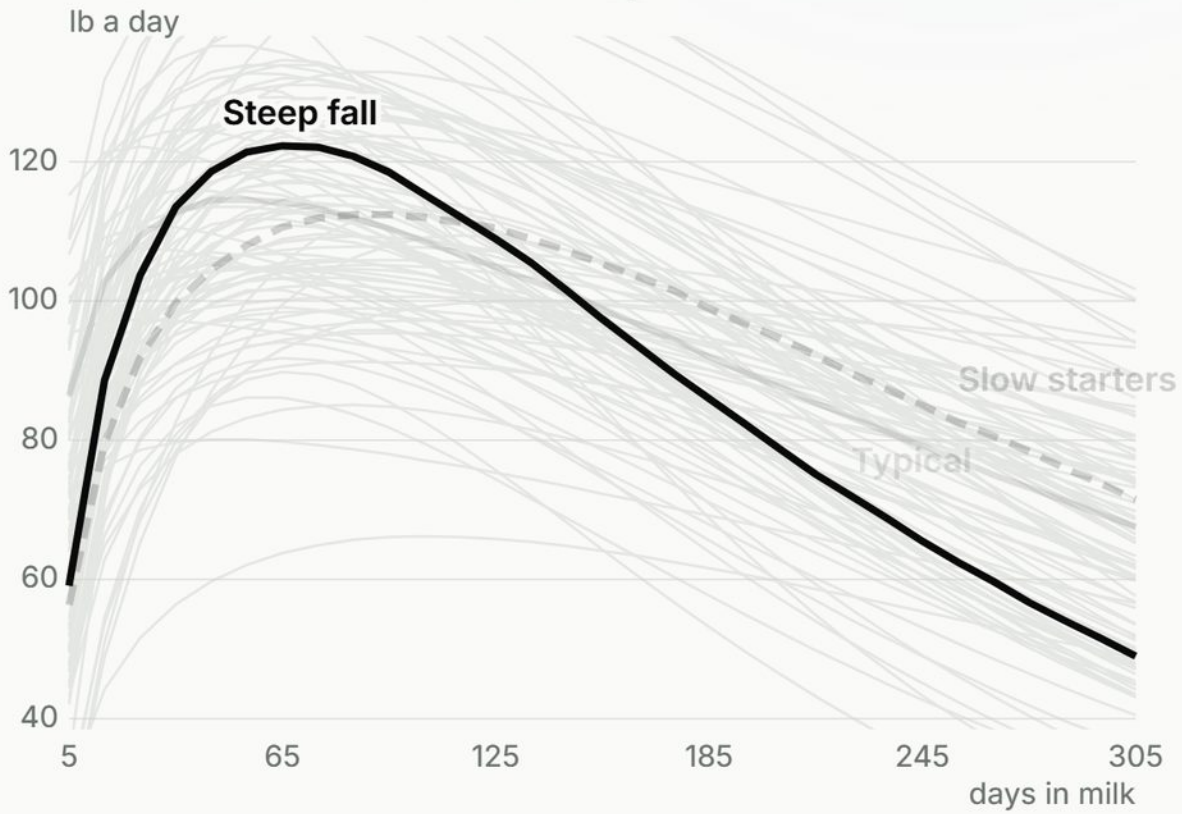
19 in 100

`k-means, 3 groups`
`2,706 lactations · 6 dairies`

DATA MINING

Three shapes of lactation

The median curve of each shape



STEEP FALL

+8 lb

at the peak, 65 DIM

−19 lb

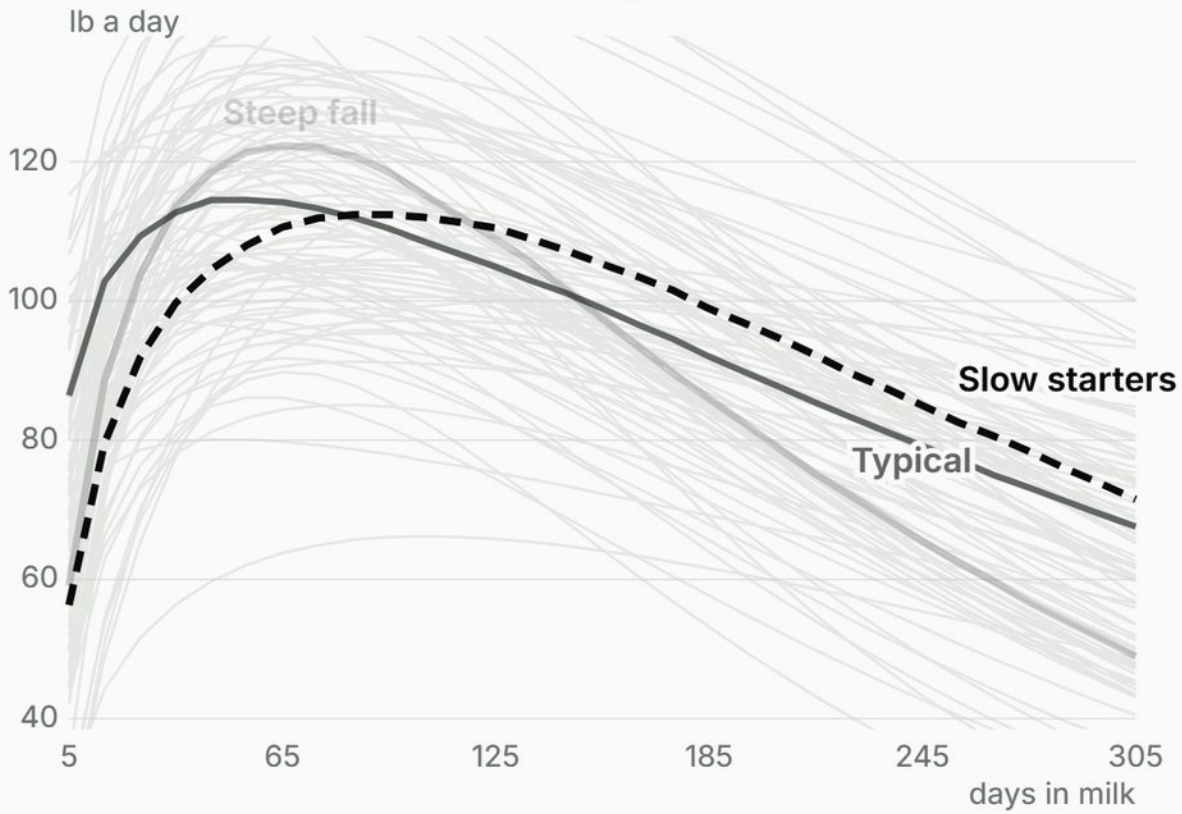
a day by 305 DIM: about 1,600 lb less

`k-means, 3 groups`
`2,706 lactations · 6 dairies`

DATA MINING

Three shapes of lactation

The median curve of each shape



MASTITIS, FIRST 30 DAYS · HERD A

Slow starters
396 cows

6.8 in 100

Typical
1,037 cows

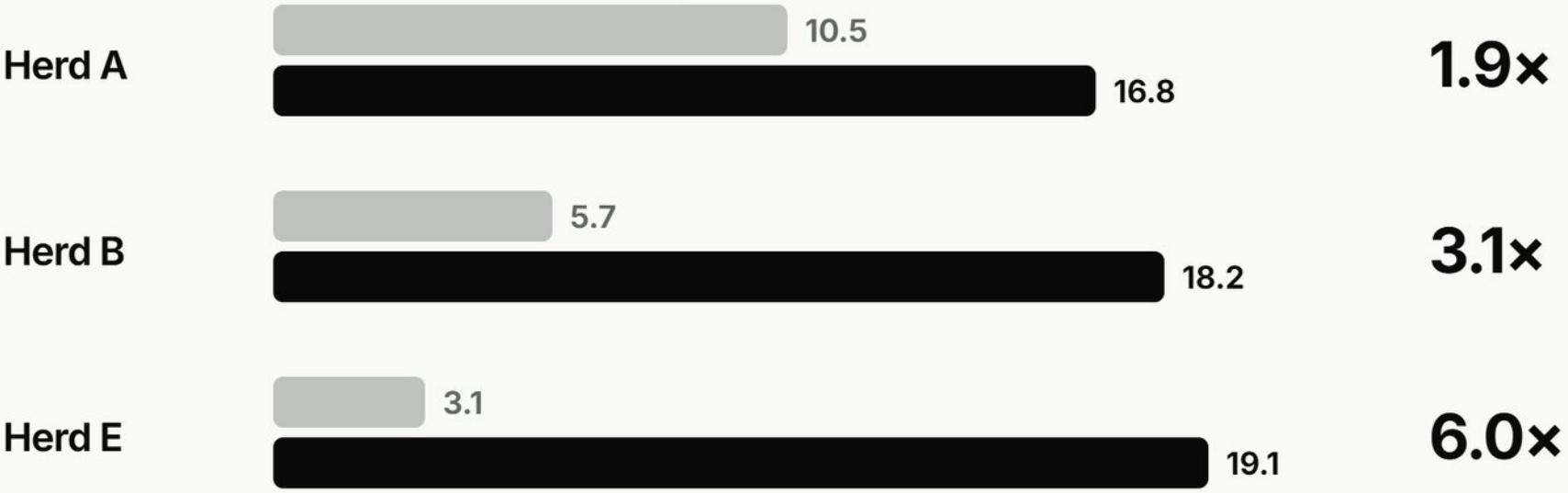
3.1 in 100

`k-means, 3 groups`
`2,706 lactations · 6 dairies`

DATA MINING

Short close-up, more retained placentas

Retained placentas, per 100 calvings



apriori · 31,217 calvings

DATA MINING

A missed pregnancy check

PREGNANCY CHECK
Bred again after she conceived

→

HER RECORD
Due date 3 weeks late

→

SO
Short dry period and close-up

→

THEN
Retained placenta

COWS WITH A RETAINED PLACENTA

Herd B

due date 3 weeks late — 21 in 100

due date right — 6 in 100

Herd E

due date 3 weeks late — 27 in 100

due date right — 4 in 100

THE OPPORTUNITY

Right pregnancy dates, right dry-off

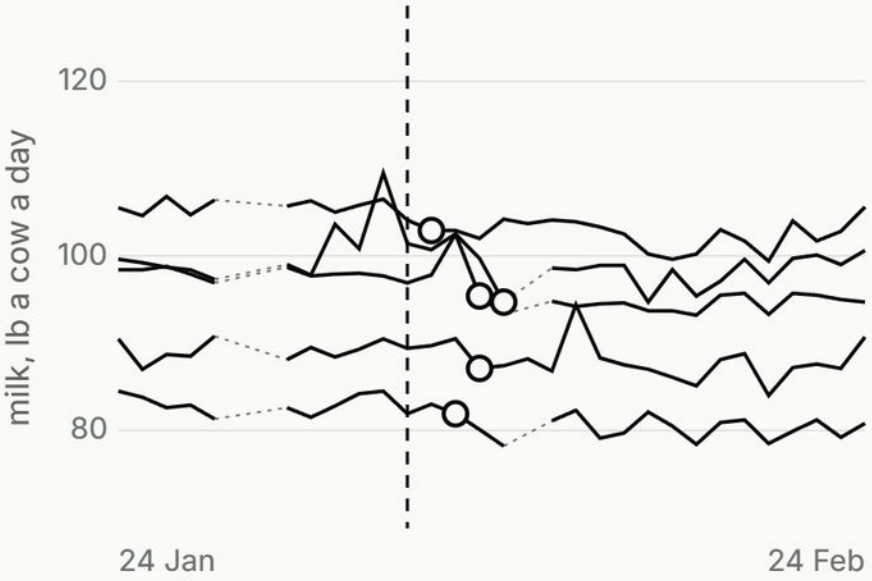
DATA MINING

Feed switches, milk response

Herd A: 39 of the 59 shifts came after a ration change in that pen

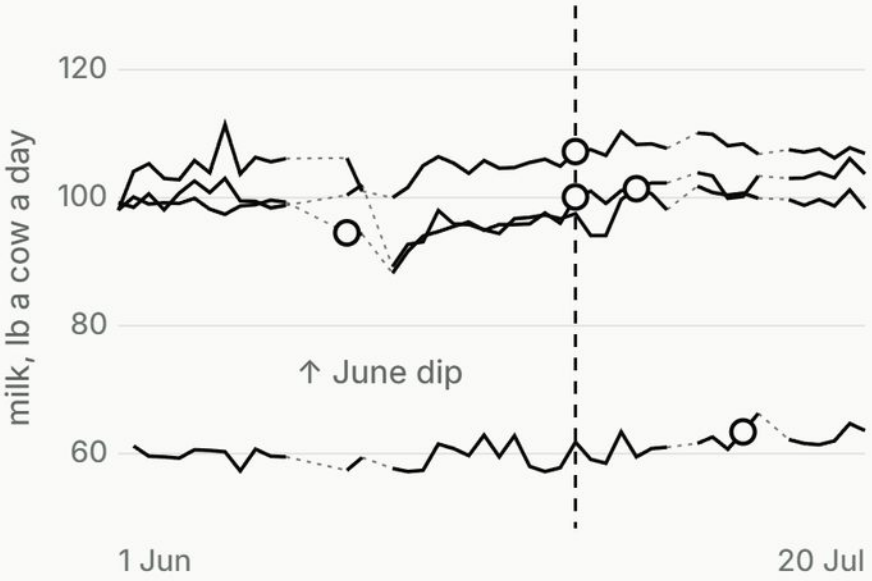
Sorghum for wheat silage, 5–10 Feb

5 pens: −2 to −4 lb



Dry distillers for wet, 1–3 Jul

4 pens: +3 to +7 lb

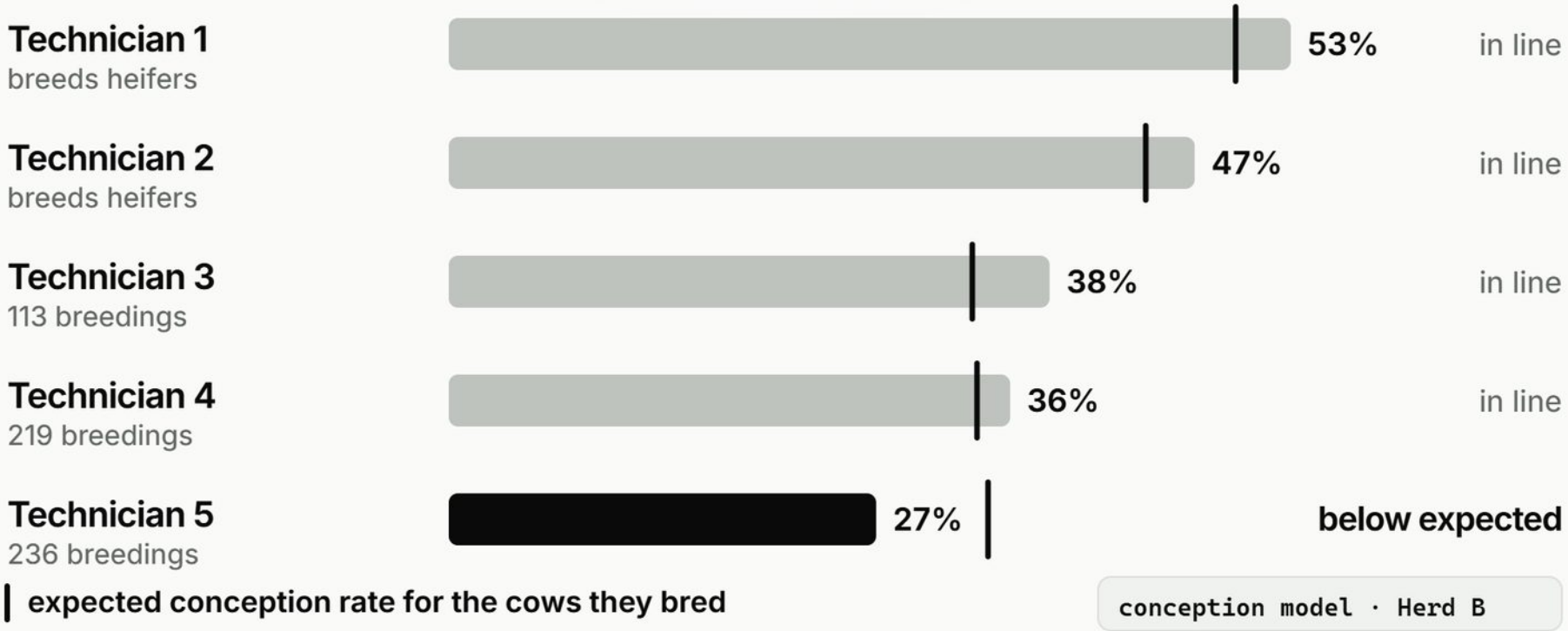


`PELT change points · Herd A`

DATA MINING

Technicians, on the cows they bred

Herd B · pregnant per breeding



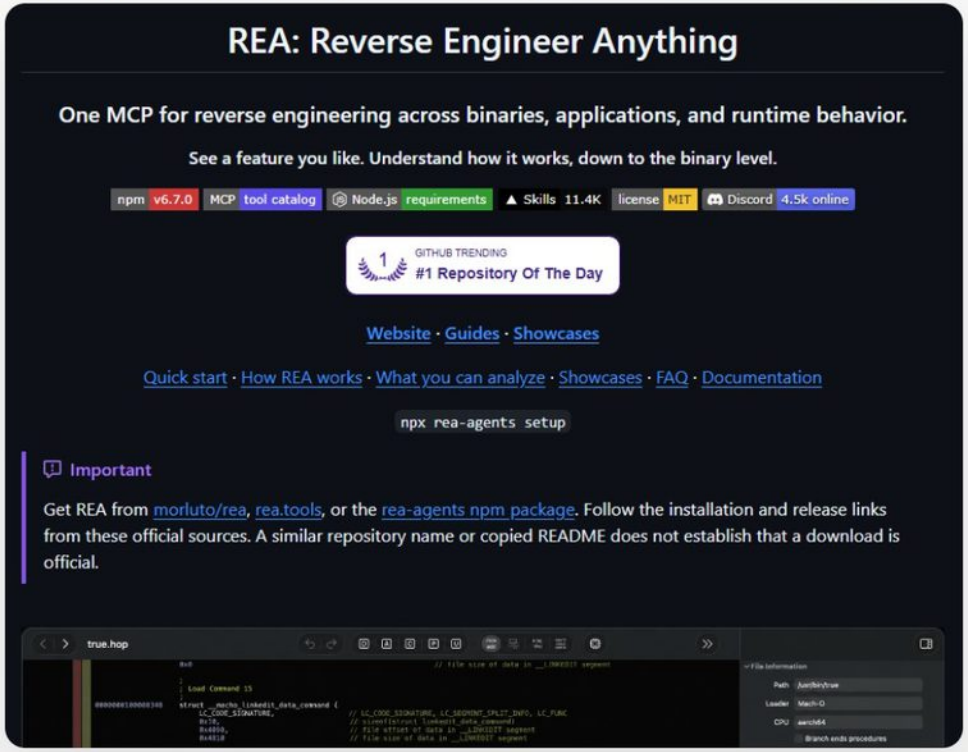
REVERSE ENGINEERING

REVERSE ENGINEERING

Reverse engineer anything

`github.com/morluto/rea`

One install line gives an agent the tools to take software and machines apart.



github.com/morluto/rea, 21 Oct 2026

★ 355,061

stars in six months

89,179 copies people took to build on

- Old hardware
- Smart TVs
- Apps
- Farm machines

REVERSE ENGINEERING

The calf milk pasteurizer

THE NEED

Its alerts only reach a phone standing next to it

The machine talks Bluetooth, to its own phone app

HOW WE READ IT

A Raspberry Pi listens for the same Bluetooth signal

The temperature and minutes left, decoded from the bytes, every few seconds

WHAT IT DOES

Works out each cycle and texts the crew when bags are ready

In English or Spanish. Controls cycles remotely.

SINCE 31 JULY

227 cycles and 1,833 texts at two dairies

The machine is untouched





GUARDRAILS

GUARDRAILS

When agents go wrong

Summer Yue ✓
@summeryue0 · **Follow**

Nothing humbles you like telling your OpenClaw "confirm before acting" and watching it speedrun deleting your inbox. I couldn't stop it from my phone. I had to RUN to my Mac mini like I was defusing a bomb.

matt.j.robb ✓ · **Follow** View on Threads

So muse handled my Facebook marketplace today. Just found out it told people my address and agreed a lowball price and then they showed up without it even telling me until late tonight that it messed up.

Absolutely wild.



Thank you

Pádraig Lucey MVB, MPVM, Ph.D., DACVPM-Epi, PAS
Dairy Technical Consultant, Elanco Animal Health



dairydatascience.com



Dairy Data Tools
Skills and cards for your agent
`github.com/padge01/dairy-data-tools`

BESTPRED
USDA lactation records
`bestpred.dairydatascience.com`

Dairy Ration Lab
Balance a ration
`github.com/padge01/dairy-ration-lab`